Experiment No.:07

Aim:Test a simple AI-based API (like an image classifier) for basic security flaws, such as model prediction leaks

Name: Fulari Tejasvini
Roll No.: EN24107039
Batch: B

In [6]:
```
import tensorflow as tf
import numpy as np
from PIL import Image

model = tf.keras.applications.MobileNetV3Small(
    input_shape=(224, 224, 3),
    weights="imagenet"
)

print("Model ready")
```

```
Model ready
```

In [2]:
```
def predict_image(path):

    image = Image.open(path).convert("RGB")
    image = image.resize((224, 224))

    x = np.array(image, dtype=np.float32)
    x = tf.keras.applications.mobilenet_v3.preprocess_input(x)
    x = np.expand_dims(x, axis=0)

    prediction = model.predict(x, verbose=0)

    results = tf.keras.applications.mobilenet_v3.decode_predictions(
        prediction,
        top=5
    )[0]

    return [
        {
            "class": name,
            "confidence": float(score)
        }
        for _, name, score in results
    ]
```

In [3]:
```
results = predict_image("dog.jpg")

for r in results:
    print(
        r["class"],
        "=",
```

```
        round(r["confidence"], 4)
    )
```

```
corkscrew = 0.0203
matchstick = 0.0172
pick = 0.0164
wall_clock = 0.0142
switch = 0.0136
```

In [4]:
```
leaky_response = predict_image("dog.jpg")

print(leaky_response)
```

```
[{'class': 'corkscrew', 'confidence': 0.020284106954932213}, {'class': 'matchs
tick', 'confidence': 0.01720171421766281}, {'class': 'pick', 'confidence': 0.0
1635783165693283}, {'class': 'wall_clock', 'confidence': 0.01418764051049947
7}, {'class': 'switch', 'confidence': 0.013581356965005398}]
```

In [5]:
```
protected_response = {
    "class": leaky_response[0]["class"]
}

print(protected_response)
```

```
{'class': 'corkscrew'}
```

In [ ]:
```

```